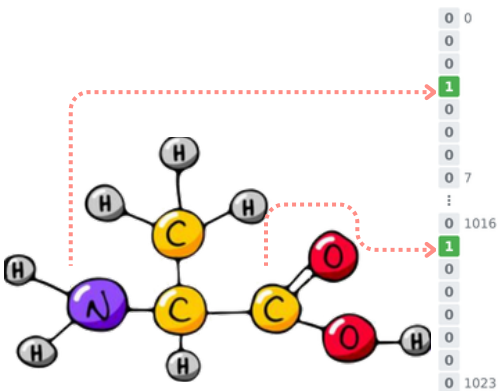


Amino Acid	1	2	3	4	...	19	20	21
1. Ala	1	0	0	0	...	0	0	0
2. Arg	0	1	0	0	...	0	0	0
3. Asn	0	0	1	0	...	0	0	0
4. Asp	0	0	0	1	...	0	0	0
...	...	...	...	...	...	...	...	...
19. Tyr	0	0	0	0	...	1	0	0
20. Val	0	0	0	0	...	0	1	0
21. Unk	0	0	0	0	...	0	0	1

Per-residue
MLPs



Substrate
MLP

	AD1	AD2	AD3	AD4	AD5
S1	AD1-S1 ↑	AD2-S1 ↓	AD3-S1 ↓	AD4-S1 ↓	AD5-S1 ↓
S2	AD1-S2 ↓	AD2-S2 ↑	AD3-S2 ↓	AD4-S2 ↓	AD5-S2 ↓
S3	AD1-S3 ↓	AD2-S3 ↓	AD3-S3 ↑	AD4-S3 ↓	AD5-S3 ↓
S4	AD1-S4 ↓	AD2-S4 ↓	AD3-S4 ↓	AD4-S4 ↑	AD5-S4 ↓
S5	AD1-S5 ↓	AD2-S5 ↓	AD3-S5 ↓	AD4-S5 ↓	AD5-S5 ↑

Positive Pair
Negative Pair

Property
prediction
head